A decorative graphic on the left side of the slide, consisting of a black crosshair with a blue square in the top-left quadrant, a red square in the bottom-left quadrant, and a yellow square in the bottom-right quadrant.

Computer Programming II (CSE2035)

Lab #11: Operator Overloading

Prof. Jaeseung Choi
Dept. of Computer Science and Engineering
Sogang University

Homework 11: Operator Overloading (5pt)

- **We will overload operators for our custom string class**
 - Adopted from our previous Week #5 Lab (HW 5-1 MyString class)
 - The class is partially defined in string.hpp (properties, constructor, destructor, and Print method are defined)

```
class String {
private:
    int len;
    char *arr;
public:
    String(const char *s) { ... }
    ~String() { ... }
    void Print() { ... }
    char& operator[](int idx);
    String& operator=(String &s);
};
```

Your job is to define these two operators in string.cpp

Homework 11: Operator Overloading (5pt)

- Check the constructor code to understand how we store characters in this object
 - Dynamically allocate heap memory to store 'len' characters
 - We don't have to store additional NULL character to indicate the end

```
...
private:
    int len;
    char *arr;
public:
    String(const char *s) {
        len = strlen(s);
        arr = new char[len];
        for (int i = 0; i < len; i++) {
            arr[i] = s[i];
        }
    }
    ...
```

Homework 11: Operator Overloading (5pt)

- **First, let's implement the index operator, "[]"**
 - **Hint:** Compiler converts "obj[idx]" into "obj.operator[](idx)"
 - After defining operator[] in our String class, we can use the code below to read or update a character of a string
 - **Spec:** If 'idx' has an invalid value, print an error message and exit

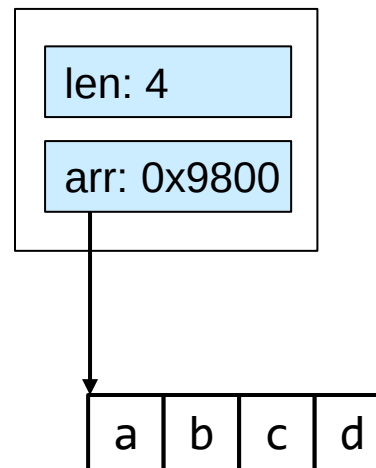
```
void test1() {
    String s1("abcd");
    cout << s1[1] << endl; // Should print "b\n"
    s1[2] = 'x';
    s1.Print(); // Should print "abxd"
    cout << s1[-1] << endl; // Should print "Error\n" and exit.
}
```

Homework 11: Operator Overloading (5pt)

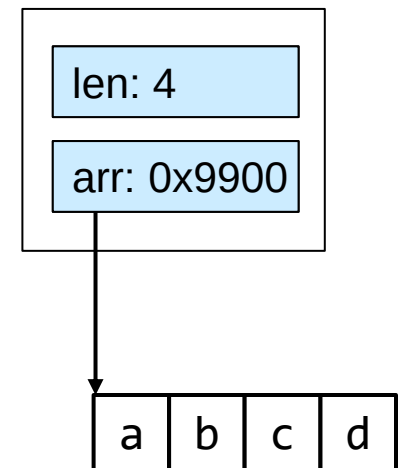
- Next, let's implement the assignment operator, "="
 - After running the "s2 = s1;" line, s2's "arr" must be pointing at a fresh memory space (*this is called a "deep copy"*)
 - After copying, updates on s1 should not affect s2
 - Make sure that no **memory leak** and **segmentation fault** occurs!

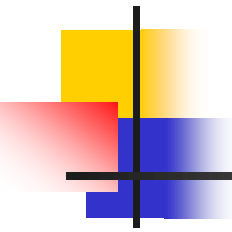
```
String s1("abcd");  
String s2("efg");  
s2 = s1;
```

Object s1



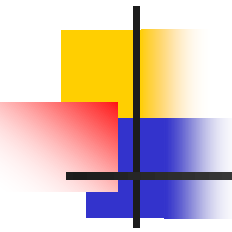
Object s2



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Submission Guideline

- **Submit the completed `string.cpp` file to “Lab #11 Assignment” in cyber campus**
 - You can fix `main.cpp` to test your code, but don't submit it
 - Do not zip the directory, just upload this file directly
 - **Please do not change the file name**
 - If the submitted file doesn't compile with the "make" command, cannot give you any point
- **Deadline is 12/16 (Fri) 23:59**
- **No late submission and resubmission this time!**
- **All the solutions will be uploaded around 12/18**

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Q & A Session
